

Translated

From: e-maxx.ru

Gray code

Gray code is a binary numeral system where two successive values differ in only one bit.

For example, the sequence of Gray codes for 3-bit numbers is: 000, 001, 011, 010, 110, 111, 101, 100, so $G(4) = 6$.

This code was invented by Frank Gray in 1953.

Finding Gray code

Let's look at the bits of number n and the bits of number $G(n)$. Notice that i -th bit of $G(n)$ equals 1 only when i -th bit of n equals 1 and $i + 1$ -th bit equals 0 or the other way around (i -th bit equals 0 and $i + 1$ -th bit equals 1). Thus, $G(n) = n \oplus (n \gg 1)$:

```
int g (int n) {  
    return n ^ (n >> 1);  
}
```

Finding inverse Gray code

Given Gray code g , restore the original number n .

We will move from the most significant bits to the least significant ones (the least significant bit has index 1 and the most significant bit has index k). The relation between the bits n_i of number n and the bits g_i of number g :

$$\begin{aligned} n_k &= g_k, \quad n_{k-1} = g_{k-1} \oplus n_k = g_k \oplus g_{k-1}, \quad n_{k-2} = \\ &= g_{k-2} \oplus n_{k-1} = g_k \oplus g_{k-1} \oplus g_{k-2}, \quad n_{k-3} = g_{k-3} \oplus n_{k-2} = \\ &= g_k \oplus g_{k-1} \oplus g_{k-2} \oplus g_{k-3}, \quad \vdots \end{aligned}$$

The easiest way to write it in code is:

```
int rev_g (int g) {  
    int n = 0;  
    for (; g; g >>= 1)  
        n ^= g;
```